Arbeitsbericht

- Name: Denis Ermurachi
- Datum: 05.10.2026
- Laborübung: Übung_03-Standortvernetzung_IPv6
- Fach: SYTD
- Klasse: 4AHITS

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1. Statische Routen:

   Am Router R-Br1 nu R-Sd1 werden folgende statische Routen benötigt, um die Kommunikation zwischen den beiden Standorten zu ermöglichen.

R-Br1:

```
ipv6 route 2001:db8:3::/64 2001:DB8:ACAD:2::2
```

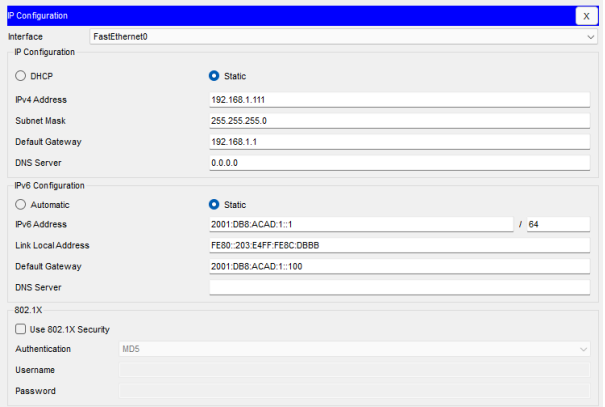
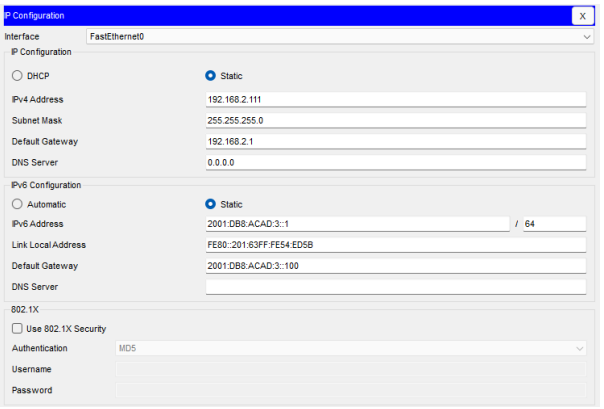
R-Sd1:

```
ipv6 route 2001:DB8:ACAD:1::/64 2001:DB8:ACAD:2::1
```

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2. Konfigurationen:

PCs

PC-A	PC-B
	

Switches

	S-Br1	S-Sd1
Vlan1 Ipv6	Vlan1 [up/up] FE80::205:5EFF:FE57:92A1 2001:DB8:ACAD:1::10	Vlan1 [up/up] FE80::202:4AFF:FE01:E864 2001:DB8:ACAD:3::10
Vlan1 Interface	S-Br1#show int vlan1 Vlan1 is up, line protocol is up Hardware is CPU Interface, address is 0005.5e57.92a1 (bia 0005.5e57.92a1) Internet address is 192.168.1.2/24 MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set ARP type: ARPA, ARP Timeout 04:00:00 Last input 21:40:21, output never, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0 Queueing strategy: fifo Output queue: 0/40 (size/max) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 0 bits/sec, 0 packets/sec 1682 packets input, 530955 bytes, 0 no buffer Received 0 broadcasts (0 IP multicast) 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored 563859 packets output, 0 bytes, 0 underruns 0 output errors, 23 interface resets 0 output buffer failures, 0 output buffers swapped out	S-Sd1#show interfaces Vlan1 Vlan1 is up, line protocol is up Hardware is CPU Interface, address is 0002.4a01.e864 (bia 0002.4a01.e864) Internet address is 192.168.2.2/24 MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set ARP type: ARPA, ARP Timeout 04:00:00 Last input 21:40:21, output never, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0 Queueing strategy: fifo Output queue: 0/40 (size/max) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 0 bits/sec, 0 packets/sec 1682 packets input, 530955 bytes, 0 no buffer Received 0 broadcasts (0 IP multicast) 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored 563859 packets output, 0 bytes, 0 underruns 0 output errors, 23 interface resets 0 output buffer failures, 0 output buffers swapped out

Routers

	R-Br1	R-Sd1
Ipv6 Adressen	R-Br1#show ipv6 int brief FastEthernet0/0 [up/up] FE80::201:63FF:FE84:9401 2001:DB8:ACAD:1::100 FastEthernet0/1 [up/up] FE80::201:63FF:FE84:9402 2001:DB8:ACAD:2::1 Vlan1 [administratively down/down] unassigned R-Br1#	R-Sd1#show ipv6 int brief FastEthernet0/0 [up/up] FE80::209:7CFF:FE1E:6032 2001:DB8:ACAD:3::100 FastEthernet0/1 [up/up] FE80::250:FFF:FE0D:4B81 2001:DB8:ACAD:2::2 Vlan1 [administratively down/down] unassigned R-Sd1#
Interface FE0/0	R-Br1#show int f0/0 FastEthernet0/0 is up, line protocol is up (connected) Hardware is Lance, address is 0001.6384.9401 (bia 0001.6384.9401) Internet address is 192.168.1.1/24 MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Full Duplex, 100Mbps, link type is auto, media type is RJ45 ARP type: ARPA, ARP Timeout 04:00:00, Last input 00:00:08, output 00:00:05, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0 (size/max/drops); Total output drops: 0 Queueing strategy: fifo Output queue :0/40 (size/max) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 4 bits/sec, 0 packets/sec 334 packets input, 22512 bytes, 0 no buffer Received 0 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort 0 input packets with dribble condition detected 233 packets output, 17932 bytes, 0 underruns 0 output errors, 0 collisions, 1 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out	R-Sd1#show int f0/0 FastEthernet0/0 is up, line protocol is up (connected) Hardware is Lance, address is 0009.7c1e.6032 (bia 0009.7c1e.6032) Internet address is 192.168.2.1/24 MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Full Duplex, 100Mbps, link type is auto, media type is RJ45 ARP type: ARPA, ARP Timeout 04:00:00, Last input 00:00:08, output 00:00:05, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0 (size/max/drops); Total output drops: 0 Queueing strategy: fifo Output queue :0/40 (size/max) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 4 bits/sec, 0 packets/sec 84 packets input, 5772 bytes, 0 no buffer Received 0 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort 0 input packets with dribble condition detected 122 packets output, 9280 bytes, 0 underruns 0 output errors, 0 collisions, 1 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out
Interface FE0/1	R-Br1#show int f0/1 FastEthernet0/1 is up, line protocol is up (connected) Hardware is Lance, address is 0001.6384.9402 (bia 0001.6384.9402) Internet address is 10.0.0.1/30 MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Full Duplex, 100Mbps, link type is auto, media type is RJ45 ARP type: ARPA, ARP Timeout 04:00:00, Last input 00:00:08, output 00:00:05, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0 (size/max/drops); Total output drops: 0 Queueing strategy: fifo Output queue :0/40 (size/max) 5 minute input rate 4 bits/sec, 0 packets/sec 5 minute output rate 4 bits/sec, 0 packets/sec 119 packets input, 7864 bytes, 0 no buffer Received 0 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort 0 input packets with dribble condition detected 187 packets output, 12864 bytes, 0 underruns 0 output errors, 0 collisions, 1 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out	R-Sd1#show int f0/1 FastEthernet0/1 is up, line protocol is up (connected) Hardware is Lance, address is 0050.0f0d.4b81 (bia 0050.0f0d.4b81) Internet address is 10.0.0.2/30 MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Full Duplex, 100Mbps, link type is auto, media type is RJ45 ARP type: ARPA, ARP Timeout 04:00:00, Last input 00:00:08, output 00:00:05, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0 (size/max/drops); Total output drops: 0 Queueing strategy: fifo Output queue :0/40 (size/max) 5 minute input rate 4 bits/sec, 0 packets/sec 5 minute output rate 4 bits/sec, 0 packets/sec 197 packets input, 13668 bytes, 0 no buffer Received 8 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort 0 input packets with dribble condition detected 120 packets output, 7944 bytes, 0 underruns 0 output errors, 0 collisions, 1 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out
Routes	R-Br1#show ipv6 route IPv6 Routing Table - 6 entries Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP U - Per-user Static route, M - MIPv6 I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2 ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2 D - EIGRP, EX - EIGRP external C 2001:DB8:ACAD:1::/64 [0/0] via FastEthernet0/0, directly connected L 2001:DB8:ACAD:1::100/128 [0/0] via FastEthernet0/0, receive C 2001:DB8:ACAD:2::/64 [0/0] via FastEthernet0/1, directly connected L 2001:DB8:ACAD:2::1/128 [0/0] via FastEthernet0/1, receive S 2001:DB8:ACAD:3::/64 [1/0] via 2001:DB8:ACAD:2::2 L FF00::/8 [0/0] via Null0, receive	R-Sd1#show ipv6 route IPv6 Routing Table - 6 entries Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP U - Per-user Static route, M - MIPv6 I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2 ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2 D - EIGRP, EX - EIGRP external S 2001:DB8:ACAD:1::/64 [1/0] via 2001:DB8:ACAD:2::1 C 2001:DB8:ACAD:2::/64 [0/0] via FastEthernet0/1, directly connected L 2001:DB8:ACAD:2::2/128 [0/0] via FastEthernet0/1, receive C 2001:DB8:ACAD:3::/64 [0/0] via FastEthernet0/0, directly connected L 2001:DB8:ACAD:3::100/128 [0/0] via FastEthernet0/0, receive L FF00::/8 [0/0] via Null0, receive

3. Konnektivitätstest:

von - zu	ipv6	ipv4
ping PC-A ↓ S-Br1	C:\>ping 2001:DB8:ACAD:1::100 Pinging 2001:DB8:ACAD:1::100 with 32 bytes of data: Reply from 2001:DB8:ACAD:1::100: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:1::100: bytes=32 time=1ms TTL=255 Reply from 2001:DB8:ACAD:1::100: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:ACAD:1::100: bytes=32 time<1ms TTL=255 Ping statistics for 2001:DB8:ACAD:1::100: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 1ms, Average = 0ms	C:\>ping 192.168.1.1 Pinging 192.168.1.1 with 32 bytes of data: Reply from 192.168.1.1: bytes=32 time<1ms TTL=255 Reply from 192.168.1.1: bytes=32 time<1ms TTL=255 Reply from 192.168.1.1: bytes=32 time<1ms TTL=255 Reply from 192.168.1.1: bytes=32 time<1ms TTL=255 Ping statistics for 192.168.1.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms
ping PC-A ↓ PC-B	C:\>ping 2001:DB8:ACAD:3::1 Pinging 2001:DB8:ACAD:3::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:3::1: bytes=32 time<1ms TTL=126 Reply from 2001:DB8:ACAD:3::1: bytes=32 time<1ms TTL=126 Reply from 2001:DB8:ACAD:3::1: bytes=32 time=7ms TTL=126 Reply from 2001:DB8:ACAD:3::1: bytes=32 time=1ms TTL=126 Ping statistics for 2001:DB8:ACAD:3::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 7ms, Average = 2ms	C:\>ping 192.168.2.111 Pinging 192.168.2.111 with 32 bytes of data: Reply from 192.168.2.111: bytes=32 time=10ms TTL=126 Reply from 192.168.2.111: bytes=32 time<1ms TTL=126 Reply from 192.168.2.111: bytes=32 time<1ms TTL=126 Reply from 192.168.2.111: bytes=32 time=8ms TTL=126 Ping statistics for 192.168.2.111: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 10ms, Average = 4ms
ping PC-B ↓ PC-A	C:\>ping 2001:db8:acad:1::1 Pinging 2001:db8:acad:1::1 with 32 bytes of data: Reply from 2001:DB8:ACAD:1::1: bytes=32 time<1ms TTL=126 Reply from 2001:DB8:ACAD:1::1: bytes=32 time<1ms TTL=126 Reply from 2001:DB8:ACAD:1::1: bytes=32 time<1ms TTL=126 Reply from 2001:DB8:ACAD:1::1: bytes=32 time=3ms TTL=126 Ping statistics for 2001:DB8:ACAD:1::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 3ms, Average = 0ms	C:\>ping 192.168.1.111 Pinging 192.168.1.111 with 32 bytes of data: Reply from 192.168.1.111: bytes=32 time=1ms TTL=126 Reply from 192.168.1.111: bytes=32 time<1ms TTL=126 Reply from 192.168.1.111: bytes=32 time=1ms TTL=126 Reply from 192.168.1.111: bytes=32 time<1ms TTL=126 Ping statistics for 192.168.1.111: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 1ms, Average = 0ms
tracert PC-A ↓ S-Sd1	C:\>tracert 2001:DB8:ACAD:2::2 Tracing route to 2001:DB8:ACAD:2::2 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 2001:DB8:ACAD:1::100 2 0 ms 0 ms 0 ms 2001:DB8:ACAD:2::2 Trace complete.	C:\>tracert 10.0.0.2 Tracing route to 10.0.0.2 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 192.168.1.1 2 0 ms 0 ms 0 ms 10.0.0.2 Trace complete.
tracert PC-A ↓ PC-B	C:\>tracert 2001:DB8:ACAD:3::1 Tracing route to 2001:DB8:ACAD:3::1 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 2001:DB8:ACAD:1::100 2 0 ms 1 ms 0 ms 2001:DB8:ACAD:2::2 3 0 ms 0 ms 0 ms 2001:DB8:ACAD:3::1 Trace complete.	C:\>tracert 192.168.2.111 Tracing route to 192.168.2.111 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 192.168.1.1 2 0 ms 2 ms 0 ms 10.0.0.2 3 0 ms 0 ms 0 ms 192.168.2.111 Trace complete.
tracert PC-B ↓ PC-A	C:\>tracert 2001:db8:acad:1::1 Tracing route to 2001:db8:acad:1::1 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 2001:DB8:ACAD:3::100 2 0 ms 0 ms 1 ms 2001:DB8:ACAD:2::1 3 0 ms 0 ms 0 ms 2001:DB8:ACAD:1::1 Trace complete.	C:\>tracert 192.168.1.111 Tracing route to 192.168.1.111 over a maximum of 30 hops: 1 0 ms 1 ms 0 ms 192.168.2.1 2 0 ms 1 ms 0 ms 10.0.0.1 3 1 ms 0 ms 0 ms 192.168.1.111 Trace complete.